Transcript: Aptiv Active Safety Teach In

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CORPORATE SPEAKERS

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PARTICIPANTS

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PRESENTATION

Elena Rosman:

So, I have just a few slides to run through. I really appreciate everyone coming this morning to our Active Safety Teach-In. We've broken today's event into two distinct sections, starting with the external perspective.

As Brian referenced, BCG is here to talk about the market outlook, and we have a special guest, David Strickland, who will give us some perspective on the regulatory environment, before we turn to Aptiv, our technology portfolio and how we're commercializing our portfolio to take advantage of the significant ramp in democratization that we're seeing take hold in the industry.

Lastly, we'll finish off with an interactive panel Q&A, so you have an opportunity to ask as many questions as you want. We understand this is a relevant topic given where the business has been and is going in terms of its growth trajectory.

Moving to the next slide, our forward-looking statement. Our presentation today reflects Aptiv's view of our future financial performance, which may be materially different from our actual financial performance, for reasons that we cite in our Form 10-K and other SEC filings, and we would ask that you interpret them in that light.

Turning to today's discussion, why are we here? For those of you who know Aptiv, we really love to do these deep-dives into key areas of our business. And this is a market, active safety, that we've been studying for a very long time.

And if you'll forgive me for being just a tad bit bold, I think, we've absolutely nailed it. We've got the right portfolio at the right time. You're going to hear that reinforced from our presenters today both from the external presenters as well as Glen De Vos.

But before we get into that, I think we'll just take a minute to remind everyone how we got here, as an industry. Most of you are familiar with this data, but the automotive industry has been absolutely uncompromising relative to safety and the improvements we've seen over several decades starting with basic content like seatbelts and airbags.

The industry has been successful at reducing the number of traffic accidents and fatalities, but you'll see recent data showing trends have started to reverse and go to the wrong direction.

And the reality is that the passive safety is no longer enough to keep people safe and we have to stop hitting things. Therefore, early active safety functionality like forward collision warning systems and blind spot detection, that are really information-only, really level zero, is just the beginning to solving this dilemma. The real benefits come from the true automation that starts with level one.

Looking at it from a global perspective, China had roughly 70% more vehicle fatalities than the U.S. with less than half of the number of miles driven on an annual basis.

And obviously, everyone is familiar with the statistics -- globally 1.3 million people were killed last year in vehicle accidents. I think one of the most shocking interpretations for me is that equates to two people dying every minute in a traffic fatality around the world.

So, with that, I think we understand the problem. Let's talk about what we're doing about it.

Today you're going to hear from Xavier Mosquet. He's a Senior Partner and Managing Director of The Boston Consulting Group's Global Automotive Practice where he has been head there for the last eight years. He actually opened up the Detroit office.

We think highly of the work that Xavier and BCG do, helping Aptiv and a number of industry participants, understand the market and sharpen our view on things like customer adoption, consumers' willingness to pay, and other market trends. You're going to hear more on that from Xavier in just a few minutes.

I think you'll get a sense that they're informed. And when we think about the potential size of the Aptiv safety market, their analysis suggests a bias to the upside.

Next you'll hear from David Strickland. Many of you know David from his prior role at NHTSA. Currently, David is Partner at Venable's Regulatory Group where he focuses on transportation policy and consumer protection, amongst other important legislative and government issues.

I'd like to also point out that David is a member of Aptiv's technology council – a group of senior technology and business leaders that informing our company on emerging trends in technology, that help guide our product strategies and the investments we make internally and externally.

And finally, Glen De Vos is here with us today to represent Aptiv. Glen serves as Chief Technology Officer, as well as President of our Mobility and Services Group.

Glen is going to walk you through how we're perfectly positioned in the active safety market for outperformance, and how our leadership position leveraged in autonomous vehicles.

Again, thank you for your participation today. There's absolutely no better time to be having this conversation, as we're seeing the penetration of active safety start to inflect – otherwise what we're referencing as the democratization of active safety. The consumers' awareness of the technology, coupled with their willingness to pay, and building upon the efficacy of the data to prove the real benefit, which David will talk more about, it is absolutely influencing the regulatory landscape.

And with that, Aptiv is exceptionally well-positioned to expand market share. Given where we sit today, we'll have roughly $1 billion of active safety revenue in 2018. That's on a trajectory to grow to over $2 billion by 2022, as our bookings continue to validate our leadership in the market.

So, with that, I will hand the stage over to Xavier.

Xavier Mosquet:

Thank you. Great. So, let me share a few figures with you and some other market trends and what we think is going to happen in the coming years.

First of all, I think Elena mentioned the severity and the number accidents. This is just the snapshot of the U.S., just the U.S. last year. 6 billion cars were involved in an accident, sorry 6 million accidents happened last year, 2.4 million people were injured, 40,000 people died unfortunately in those accidents and 13 million cars were involved in those accidents.

This is only the reported figures, and in the last figure, which is the number of cars and the number of crashes, we know for sure and it's on those figures that this is higher actually than the figures that have been reported.

Those figures are pretty bad in themselves, but the thing is, actually, they are getting worse. So, in 2014, we only had 33,000 people who died in accidents and we have 40,000 this year. It's a convergence of two things.

One, yes, the number of miles have increased due to the economic recovery but also, actually, the rate of accidents has been increasing. So, it's been declining for basically 30 years and now, it has started to increase again. And therefore, you're doing something is critical.

A few years ago in 2014 and 2015, the industry association commissioned BCG with a study to look at what were the ways to increase safety on the U.S. roads. And when we did that work back then, we actually looked at the number of accidents and the costs of those accidents with all available public data and the cost of accidents in the U.S. is about $950 billion, 5% of GDP every year.

So, the economic value as a society is huge and this is why, by the way, we're got to talk about regulation because as a country, as a nation, it is actually, the stakes are very high.

And so, this $950 billion was computed in 2014 and 2015 and now, it has probably increased by already 10% to 15%. Now, the solution -- I think one of the solutions is definitely active safety.

Since actually in 2010, the administration has launched a number of studies to better understand what were the benefits of active safety features on the U.S. roads. And all the results you have here are how many crashes or the percentage of crashes that can be avoided by active safety technology and these are studies that have been run somewhere between 2010 and 2015.

And you can see that rare automatic braking, you can avoid as many as 60% of backing accidents. And if I look at front-to-rear accidents, automatic emergency braking or forward collision warning can avoid anywhere between 30% and 50% of those types of accidents.

And so, we do have the means with available technology, technology that has been actually available for the past 10 years, we have the means actually to increase safety on our roads and this is what I think this meeting is about today.

So, on the work we did with the industry association, we looked at every single accident in the U.S. with data available from the insurance companies, NHTSA and other available sources. We looked at all the main active safety features that were available back then and what percentage of accidents could be avoided. If you look overall, about three or four main active safety features that are available, you can prevent 30% of existing accidents on the roads in the U.S.

And therefore, the economic benefit of available technology is about $250 billion. If you do the math, it's several thousand dollars per car in terms of added value every year. And so, the payback is obvious. The question is how do we make it happen?

And there are a few things that have to get together for this to happen. First of all, but it's not the only thing by far and that's why we're going to talk about the different factors, the regulation obviously has a role to play, but again, it is not the only one and we'll see that it may not be in the end the main driver.

Technology evolution is critical in two steps. One is the cost of those technologies and how can we make them affordable for the consumer. The second obviously is the degree of usability of those technologies. So, how often are they effective and can be used in different weather and work conditions?

Then there is a consumer awareness and the willingness to pay. Interestingly, if I look back a few years ago, a few people were even aware of those technologies and the benefits that they could have.

I still remember a couple of years ago, talking about the forward collision warning systems I had in my car for 10 years and how nobody in my family would be allowed to drive the car without that, and I would say three years ago, I mean people were looking and saying, "That's interesting. Maybe I should try."

And yes, I think unless you haven't tried, definitely, you have to try. The last factor is the OEM willingness, and that's particularly important in the U.S. because in Europe, you basically order a car and it's made for you and therefore if you want a feature, you'll likely to have it.

In the U.S., we all buy cars from the lots and therefore if the feature is not available, we're unlikely to buy it. And so, the way that OEM engineers in a way purchase and precondition these cars, for us, it makes a huge difference.

And interestingly, if you look back, two or three years ago, the differential in penetration of those features in Europe versus the U.S. was essentially lower in the U.S. and one of the main reasons is just product availability.

So, if we look at regulation, I wouldn't spend a lot of time on this very crowded chart but what you should know is that in Europe, active safety has been introduced in the way that the regulator looked at the Euro NCAP so how cars are rated for safety has been there for several years.

And so, the real Europe administration has moved from purely passive safety, I resist an accident, to how can I avoid an accident several years ago. Even today, now, it is already your part of the plan for the U.S. administration, but even today, we don't have yet active safety in the formal application of the rating and administration.

So, a few years ago, the administration started to investigate and to start to recommend a number of these active safety features, but were only at the start of the real regulatory process in the U.S. So, we know it's going to happen, we know how it's going to happen, but so far, it is not yet in our ratings.

We know for sure then that the five-star rating will have to include active safety to be available in the U.S. So, this is changing and therefore, some of the market trends that you're seeing today, actually are only the start of the growth of that market.

There are a few things also that happened and I think the administration has done an incredibly good work. First of all, even without regulating, is that we actually gathered all the OEMs to decide together on a voluntary plan outside the context of legislation.

And 20 OEMs in the U.S. have agreed to make automated emergency braking standard in their cars in 2022. So, that's definitely a revolution. I would say even in Europe where the regulator is ahead of the curve, this has not happened yet.

The second thing is now, it's been clear for everybody that AEB will be part of not only the recommended list of technologies but will be part of the five-star rating in the coming years so that's something that all the OEMs now are preparing for .

And then as I mentioned, there's a number of studies several than that have been launched to better understand the actual societal benefits of your various active safety technologies. These are typically studies that are taking three, four, five years to give the results but a lot of them are now coming up and showing the incredible value that we can have for all of us.

So, this is just to give you a snapshot of what's happening in Europe. In Europe, you'll be automated, sorry, the active safety features have been part of the waiting of the safety features for now quite a while.

The formula is set. Everybody knows what it is and all the OEMs are complying to it now. And if you look at some of the roadmaps, what's important here I think is the red dots. The red dots are the time where OEMs have to comply in the marketplace for actual sales.

You will see that AEB, it's automated emergency braking, for it's latest version will be mandated in 2018 as part of the safety rating. And then in 2020, most active safety features will be part of the official roadmap and the safety rating for all the OEMs in Europe. And for us, as I said in U.S., this is only the start of that process.

OEMs have embarked in this trade I think wholeheartedly. I said the 20 of them have already committed that without regulation to make in the U.S. AEB standard in 2022, but a number of them now are using that as a key differentiator in the marketplace.

And what you have here is just a few examples of what Volvo, Toyota, G.M. or Nissan are seeing out there. And just take one quote from Volvo, which is by 2020, no one will be killed or seriously injured in the new Volvo SUV and G.M. has now its new three sentences mandate that you may have heard before which is, zero crashes, zero emission and zero congestion. What's important to us is the zero crashes.

Both of them have committed to make AEB available. I think the Volvo figure at 68% already of take rate in AEB I think it's telling. All the OEMs are now heading to that and Nissan, which is only at 14% right now, has committed for this year to 1 million cars in the U.S. sold with AEB. Knowing that they actually sell 1.4 million cars, I mean that's close to 70% of their fleets.

And I think we're seeing out there the OEMs, especially in the advertising, using that as the next frontier of the differentiation with customers. And so, that will significantly increase the markets, one is around 1S features.

This is just to give you an idea of how the 1S is today. This is the number of consumers in the U.S. who are telling us that they are either familiar or very with some of those features.

The main one to the left is the blind spot monitoring, and that's already 50% of customers in the U.S. who said they are familiar or very familiar. These figures are already fairly high. And if you look back in three or four years ago, those figures would have been -- would have been extremely low, which mean that the advertising, the consumer information, the dealers on the lot and worked with us are now making a big difference.

So, this actually results in a significant market growth. If you look back in 2014, where very few OEMs and customers were talking about these features, I would even say even the administration was not totally convinced that it was critical.

This gives you an idea of the growth rates just in two years. What you have here in dull green is the penetration of those features in 2015. And the top bar is the penetration in 2017. So, you can see that in just two years, the market, the penetration rate in the U.S. has doubled.

And as I said, this is actually was the warning about regulation but no real regulation yet and so you can expect those figures to continue growing at the same pace for the next few years. So, a huge increase in just a few years where this was an unknown entity in 2014 and now to significant market increase.

And a few things that I think are supporting that is the real consumer willingness to pay. I mean typically, as you may know, consumer doesn't pay for safety. If insurance was not compulsory then we probably not have one.

And frankly, seatbelts have taken years to come into the mind of people and if people have thought that they have to pay for it, they would probably not have taken it.

Here, it is different actually. Customers start to understand the value for them and their families and there is actual dollars that it would put behind it. And so, these are some ideas of the average willingness to pay of consumers, so obviously, some consumers would pay less but some would pay much more.

But roughly for every single of these features, the first one being forward collision warning here, people are willing to pay roughly $500 for this feature. Now, if I start bundling some of those, right now, you can buy for most OEMs a safety package that's about $1,000, $1,100 that bundles a number of those features.

And if you look at the price of those bundles, they basically are some of the some of this willingness to pay. And so, there is a real market and this is what I was saying and which is growing before regulation because the customer as they start to be educated about it wants to buy it. And you can put it on the lot. If the dealer starts to want to sell it, it is actually being sold.

The other thing that I would like to express here is the owners like it. So, what's the degree of satisfaction? Even we think that you only use in emergency situation like automated emergency braking or forward collision warning, the satisfaction is close to 70%.

And with blind spot detection, it's above 80%, and so, this is not only a start, this will continue because your consumers just like it. And as I told you, I'm personally one of those guys that will never allow anybody in my family to buy a car without it. And as soon as you have this thought in mind, for sure, you've got to continue with it.

I'm going to skip that because I'm sure you're all familiar with all the definitions of the SAE levels and those various kind of features. But what I'd like to tell you is there is actually more than the levels of the SAE.

There's actually a significant continuum especially between Level 1 and Level 3. It is not going to be a pure level story. And so, we started to invent a number of things like in Level 2+ or Level 2++ because actually, there is a difference between Level 2, which is basically, I'm fully in-charge and I have something that helps me drive, and Level 3, where I stopped to be in-charge and I know where the system does it for me.

And the differences are, is the driver in the loop and for how long, i.e. can I take my hands off the wheel for one second, three seconds, 15 seconds, 2 minutes or forever? And this, as you can see, between one second and forever, there's a huge continuum. And the technology that's behind it is radically different. And so, you will see an evolution from Level 2 to Level 3 and 4 that will be more continuous at the levels you seem to suggest.

The other thing is, can the car get to a safe stop, yes/no? It depends. But if you want to go to real safe stop and not just stop in the middle of the highway, which is some of those features you could be doing then it does require a significant set of technology. So, between 2+ and 3, there will be some significant differences.

And then is my system redundant i.e. if the system fails, is there a backup or am I the backup? And so, all these things will show that there will be a market for Level 0, 1, 2, 2+, and then there is a gap for 3, 4 and beyond because you need to add safety, backup, filtering systems that I think are not part of the usual active safety story.

What's interesting is, if you look at all these levels for which there is actual demand in the marketplace, they build on each other, i.e. this is the same type of sensors, the new one that appears somewhere in this branch is the lighter, but many of the other sensors are the same, but the number of sensors, the type of sensors evolves as we go through those levels.

And so, as the market enriches and customers ask for more, the CPV, for the industry and for the suppliers, the content per vehicle increases. And so, this is a nicely enriching market. And the other thing is -- and we've said that for many years, there will be no autonomous vehicles without the huge trend to active safety because we have to build scale in those sensors and processors to be able to make that affordable.

When we go to the Level 4 and 5, which will be expensive, we need to have that skill in active safety so that we have enough volume to reduce the cost and availability of those technologies. And so, the great thing is the market as today is paving the way for tomorrow's market, it's also paving the way for affordable cost for the autonomous technology.

Just in terms of idea, if we think we can get today 30% safety improvement just with Level 0, 1, 2 and some of the 2+, we're going to get probably to 90%. And when it gets to autonomous technology, one figure to remember is that 94% of accidents are actually human error-driven. That's the visual idea that if we can get autonomous to an affordable and right level of technology, we're close to 90% of safety levels.

So, what does it mean in terms of market? And I'm going to focus on the right-hand side here. Our view is that the market today is about $5 billion just for the active safety technologies, mainly split between L0, 1, and L2, L2+. And that market in 2022 should be roughly around $11 billion.

We think there is upside, and the upside actually would come from the mix of 0, 1, 2, 2+ and 3. And given how customers actually shifted from one level to the other to increase the value for them, we think the market could be actually slightly higher.

If I look at the base scenario, it's 17% annual growth, and so far higher than the average automotive markets, and in the higher scenario, with 21% compounded annual growth rate between now and 2022, and so, a significant increase and significant market opportunities in this area.

In summary, I think there's a few messages I would like to leave you with. One is that the societal value is huge and therefore you can expect the regulator everywhere to continue to be all over this. Now that it's clear and that there is this value, and by the way, China, for instance, has already announced that it will commit to the Euro NCAP levels. So, it's happening in Europe, it's happening in the U.S. and it's going to happen also in other large automotive geographies.

The second thing is OEMs definitely are using that as the next step in differentiation, so expect the prices decline, expect the communication to customers to be high, and at some point, it will become a must for any new car to have those technologies.

The CPV and the penetration rates will continue to grow and it's going to be a nice continuum between L0 to L3 and 4. And as I said, this is definitely paving the way a fewer down the road for L4 and L5 technology. Thank you. David?

David Strickland:

Thanks, Xavier. Good morning, everyone, and Xavier, thank you for the fantastic presentation. This is going to be the truest notion of a person who does a lot of data analytics as part of a consulting group and a former government official in terms of the slide deck. That's the only slide because this is going to be a lot of talking and hopefully leaves us some good questions.

Xavier gave a fantastic foundation for this discussion. But I do want to talk a little bit about the regulatory environment writ large in the United States and the differences with Euro NCAP and why it's so, what impact it's going to have as we move towards the stair step -- the stair steps going to the highest levels of automation, Level 4 and Level 5 from SAE, and ultimately, what this means for the safety marketplace.

Xavier did a good job in sort of talking about the numbers. Clearly, from a regulator standpoint and from a National Highway Traffic Safety Administration standpoint, we have done an excellent job in using crashworthiness and passive safety programs and behavioral programs to try to influence the numbers down.

As we've had absolute explosion in vehicle miles traveled, we have up until recently very successfully reduced the fatality rate and the crash rate. As Xavier also mentioned, in the past two years, we've seen increases of fairly significant amount, and frankly, in terms of comparing from the 1950s to the day, we've probably seen the two highest level of crashes in terms of fatalities by users we've seen since we've been keeping track.

And once again, there's not an easy reason as to why cars are safer, crashworthiness is better. People are wearing their seatbelts more. But we're seeing these increases and I think Xavier mentioned a couple of real good reasons but we've had some economic upturn from the lowest point.

Actually, when I was still in office, it's about 32,000 people approximately died in traffic cars which is the lowest that we've had like on record for that particular point because we keep track of these things which is a wonderful downward trend. And actually, from about a 10-year period from 2005 to 2015 plus or minus, that was about a 25% reduction in crash fatalities in that window, which is also a record for the United States.

And we were seeing similar things actually happening around the world in terms of crashworthiness as well. So, what happened, why do we end up losing 37,641 people in 2016, which is a significant increase of what we've seen from our lows?

I think there is a number of issues that we have a hard time quantifying, which is distraction which the agency estimates to be about 3,500 people a year dying because of distraction-related crashes. So, that's a whole spectrum of things and behavior.

But with devices really entering to the marketplace, it's really probably the one X factor because it's really hard to say, once you've had a crash, people are going to easily give up, they're going to say, "Oh, I was watching a streaming Hulu feed while I was driving," or you're not going to get that.

So, please accident reports don't pick up that type of behavior, which means it's really hard to sort of correlate the actual real number in terms of also too, discretionary driving is often the most dangerous type of driving, those trips that are not related to your normal pathways to work or school or home.

It's those longer trips out to different places, it's people actually going out enjoying life more at restaurants and drunk driving. All of these things, we don't know all the corollaries but they're there.

And the one answer that everybody is talking about in terms of the moon shot is how to get to the highest level of automation so we eliminate driver error from a driving task, have the vehicle do it all basically itself.

But before you could sort of talk about Level 4 and Level 5, you're going to have to get back to the foundations of frankly active safety. It's the things we were seeing penetrating the market days, so things that the regulator is looking at very, very hard, and how to encourage it.

And so, let's circle back to Xavier's talk, Europe has moved much quicker on these things than we have in the United States. And I'll tell you there's a couple of reasons why which are frankly going to be long-term issues, actually how you're going to get regulations on the book or even changes to the new car assessment program here in the United States.

The European model actually is much more flexible in terms of the speed of what it has to underpin on what it recommends as part of NCAP. And then it also has a much more regulatory threshold as to how to award credits for a five-star rating.

So, the European system is much more flexible and malleable. And so, which allows it to be able to bet on the come, if you will, right? They can see, frankly, not a very narrow dataset and they can make some decisions as part of the European Union for it to change their NCAP in a way that its frankly almost impossible in the United States.

In the United States, the Administrative Procedures Act, APA, and the way the regulations are formed in the United States, you need tremendous amount of data before you can even begin the process of moving towards regulations.

You have to prove cost-benefit, and with that, you have to actually have enough data to show that whatever the device or issue that's actually on the vehicle does what it says it does, and in addition to that, actually, it can prove out that benefits outweigh the costs.

And as a former NHTSA administrator who often bang my head against the wall on a number of issues which were frankly commonsensical like the backup camera rule that was finalized back in 2014, which I worked on in some various thesis for over decade, it takes that type of individual market penetration, manufacturers making decisions to put these particular systems on their vehicles. And we can talk about the history of the ad nauseam.

Probably, one of the clearest sorts of pathways is electronics. There's actually ESC, electronic speed control. That was introduced by Mercedes-Benz in 1990 as an option. And because in order to be able to address lots of vehicle lateral control and rollover and all those natures, all those things, very effective technology, but it was also on the very highest priced vehicles.

Over time, you've got some market penetration, you've got some democratization from competitive forces of the manufacturers turning to a stability control systems until you get to probably about 2003, 2004.

You had approximately 45% penetration of ESC systems throughout the fleet. And it was really showing some benefit and the agency was taking notice. In combination with a legislative mandate, that was part of a highway safety bill in 2005, electronic stability control was actually mandated by the congress because of a combination of things in terms of the agency supported that particular mandate, there was a number of issues that were going on in terms of rollover crashes which were really sort of hard to sort of nail down back then in the fleet.

But even with that congressional mandate in 2005, the actual rule, the actual [FNB] assessed for ESC did not come to pass until 2012. So, recognizing this typical trim lighting in the United States about not actually having a formal regulation on the books, and another example is rear view camera, the first rear view camera I guess study probably happened in 2002 or, I guess, rear visibility, I guess, improvement in 2002.

It became a congressional mandate once again in 2008, and it was actually finalized in 2014. So, what is the force majeure that influences American automakers and people that are selling in the European and Asian manufacturers that are selling in the United States, why do they move forward in such a way to introduce these systems and build these stair steps when it takes so long to actually get a force driving democratizing regulation which mandates every vehicle have a particular system on the car.

Well, there are two things that are there. As we note, the New Car Assessment Program or NCAP, is a very powerful tool which has influenced a number of things. Then the update that I oversaw back in 2012, which actually had the side cold test for side impact crash to other things, it is the force-mover or the market influencer, not mandatory, but improves the margin of safety.

But even with the New Car Assessment Program, it isn't something that the agency can simply do. They have to go through similar analysis for a rule, they have to go through figuring out a particular test procedure to be able to test it, to be able to harmonize it, to be able to see what's most effective and then being able to generate the notion of benefits even though it isn't a rule that you actually want to place it on the Monroney label so you can influence people to actually say, "This vehicle has this particular technology."

And which is they have the checkmarks for a particular active safety systems back then. You had the forward collision warning checkmark, you had a rearview camera checkmark eventually, which has hopefully influenced the market and actually influences manufacturers to take to endeavor into that, which is the right thing.

The second part of why these issues I think will continue to be a foundational stair step is because the agency even without a regulation pushes manufacturers very hard to try to include these things.

As they go through their data and they see particular promising technologies, the agency through it bully pulpit works with manufacturers and talks with the, frankly, even consumers in some ways and through NCAP and other programs to actually invest in these technologies and Xavier was talking about the take rates and the prices were coming down.

And that's one of the things in terms of bully pulpit that it uses and really sort of helped support the marketplace for things. And it knows it's not going to necessarily be able to mandate for four years to a decade.

The other aspect or the important issue about this is, is frankly the standard of care and safety of the marketplace for vehicles. Now, when you're thinking about democratization of technology, the agency can do a few things, as I said, New Car Assessment Programs, they could do guidance.

Most recently, the last administrator, the last confirmed administrator, Mark Rosekind, and the team put together frankly a negotiation of voluntary agreement for manufacturers to adopt automatic emergency brake.

The reason why that they did that, how do you influence the marketplace faster than the regulation to be able to take on foundational active safety systems? Automatic emergency braking is something that the agency had been working on even when I was the administrator looking at the great promise of it.

They generated enough data to frankly to say, “This is something I want to make a big bet on. We’re going to talk to manufacturers and we’re going to get a voluntary agreement in place so we can actually force this technology to the marketplace,” knowing that it is a foundational for getting in the automated driving, but also immediately improving the crash rates and the crash risks on the roads.

Hard work for them to get the voluntary agreement, and they got it. It was a little controversial because it is not typical rulemaking process where you have notice and comment and you have sunshine laws and have all these other things in terms of talking about these issues, which isn’t there when you’re talking about a negotiated agreement with the manufacturers, which makes some safety advocates uncomfortable.

But what it does do something is something that a regulation can't do without a decade’s worth of work. They’ve got the manufacturers to make an obligation so that in 2022, all vehicles should have AEB.

But with that, there’s also this notion whether it’s guidance or a voluntary agreement, which changes the marketplace and actually makes other manufacturers, competitors and participants to actually dive in to the ADAS space for active safety space standard of care.

Once a vehicle is seen or peer vehicles are seen as having a better crash performance and other issues, this is where the civil liability system, the trial bar may come into play. When you see things that are not regulatory that’s like issues like let’s not let the distraction guidelines that they issued a few years ago, they have no force of law.

The manufacturers cannot be enforced against for not following the guidance. The guidance is guidance. It’s the same thing as for the automated vehicle guidance, the same thing. It is a suggestion, it cannot be enforced.

But what happens is, if you have a vehicle, you have a system that does not follow NHTSA guidance, it does follow NHTSA’s suggestion, trial lawyers will make an argument that this in and of itself may pose an unreasonable risk to safety.

With that, we can make a decision with our friendly court in Eastern District of Texas, take your pick that, “Yes, you’re right, a vehicle without this particular system versus others may actually be proven to be an unreasonable risk for safety because we know for a fact that this type of crash could be avoided.

So, that’s out of the reason why once you get deeper penetration to the marketplace and you have more vehicles on the road with these systems, you see other manufactures that may not have necessarily made those bets or investments or research earlier diving in and adopting in a faster rate.

Because at the end of the day, if you end up with a crash scenario where one vehicle always performs better than another vehicle, you could create a situation where you may be facing legal liability for not taking on a particular safety system as soon as you should have in the eyes of the regulator of the courts.

An example of this is actually what happened regarding Jeep Cherokees a few years ago. Everybody that made SUVs probably in the early to late 90s, early 2000's, had their fuel cells directly behind the bumper and the rear part of the vehicle.

So, you had rear-end crashes which pierced the fuel cell and led to fires and in very aggressive crashes which even at lower levels of actually kinetic energy in a crash became fatal because you had a fire from a fuel cell being punctured.

So, manufacturers on their own because probably of civil liability suits and settlements, they redesigned and they moved those fuel cells more mid-vehicles so they can increase the crash, throughout the crash radius or the crash parameters around the vehicle if there's a rear-end crash.

Chrysler Jeep chose not to do that and they actually had a series of vehicle crashes and fires that were well above their peers. That vehicle was completely compliant with Federal Motor Vehicle Safety Standards at the period of time when it was originally designed, it was state-of-the-art.

After about a 10-year period, it was no longer state-of-the-art. And the standard of care had changed so that now, a vehicle was judged against its peers to be a safe vehicle and a reasonably designed vehicle became an unreasonable risk for safety which made a larger recall and a number of changes to the vehicle exposed.

So, from a legal regulatory standpoint, I know the boring part of the presentation, I don't have the sexy numbers like Xavier. Glen is going to talk about active.

But the legal environment in terms of how this influences investment and purchasing decisions is that active safety is so deeply penetrated into the marketplace at this point even without United States being on the leading edge of being promulgated, regulated and mandate these things. You will see deeper penetration just because the forces around NHTSA's pushed guidance influence in the civil marketplace.

Now, NHTSA will continue to work and they will continue to work on improving NCAP and they see the need to try to using the NCAP as a way to encourage active safety. It's going to be difficult to how active safety will be wrapped into the five-star because that's going to be the part where NHTSA, even when I was working there, I was having an issue with because, how do you prove a crash avoided.

It's an entirely new dataset. It's easy to prove a crashworthiness car got in a crash, the passenger and driver cell was well-protected, there wasn't any intrusion. You have the biofidelic dummies, of which really where I can record injuries better.

You have all these things which you may actually prove. If you do this to this car, if you improve, if you add side airbags, you will get this result, yes, there is a cost per vehicle, actually, we get this benefit of why.

But if you put on a sensor suite which has a significant cost, how do you prove that this crash was avoided and how do you enumerate that? And that's going to be one of the difficult things for the regulator and ultimately the Office of Management and Budget to figure out how do you quantify this stuff and can you actually quantify the number of when it crashes?

Probably, the highest level of this automation, you can probably figure it out because you have the cameras, you have everything you could see, like that was clearly an unavoidable crash and we can tick that.

But at Levels 1, 2, 2+, 2++, much harder to figure out, but that's the reason why the agency us using all of these leverages and triggers in order to be able to influence the marketplace in different ways to be able to get these out there and then the pure numbers going down with the evidence of crashes avoided versus an individual car by car analysis which they use using the Fatality Analysis Reporting System.

Or the FAR system or the [Guess] system, all the things that they use to sort of quantify numbers. And NHTSA is actually in the process of updating this data modernization project to, once again, help sort of catch and quantify these things.

So, in conclusion and really sort of the takeaway is that the United States, the National Highway Traffic Safety Administration, the Department of Transportation regardless of political stripe are fully committed to the stair steps back to safety.

They recognize that this is a real way to improve crashes and even which was a bit of a data laggard lane keep assist because I know the Insurance Institute for Highway Safety was saying, they were seeing actually a few years ago no benefit in lane keep assist because people found it annoying and they cut it off and there's other things.

But then actually even with that, the Insurance Institute has now found that there is a benefit. They're actually picking it up now, that the agency is willing to once again bet on the combat, the only way we're going to get crashes down well below where they currently are is by having the vehicle take over more of the interdiction task of having active safety being involved.

And I think what the current marketplace as stated and what the legal environment stated, you're only going to see more and more of these systems being adopted even by those laggards in the market. They necessarily have made the investments because it's going to be a force majeure in terms of how vehicles perform in the fleet versus themselves.

And nobody wants, if there's anything manufacturers were little by the schooling fish, right? You may not necessarily want to be the first fish in the line because that's a lot of cost, but you definitely don't want to be the last fish either because you're going to get eaten. You kind of want to be that comfortable middle and that comfortable middle is frankly was where the marketplace may be moving everyone to in terms of how these systems work.

Well, I will conclude with that looking forward to questions and answers. Nice to see so many familiar faces that I've seen from a number of other events, and thank you so much, we appreciate it.

Glen De Vos:

Okay. So, good morning. It's great to be here to have the time to talk to you about active safety. As you saw with Xavier's and David's material, we're really at an inflection point in the market. What I'll cover today is our take on how we're positioning Aptiv relative to that change in the market dynamics, how we want to take advantage of that, and leverage that to grow our business.

It is interesting to see what's happening now. When I joined, for those of you who don't know, it was at the time Delco, which was part of Hughes Electronics back in 1992.

And that was right about when radar was first being thought of as potentially applying to automotive for backup aids and things like that. When you think about the ensuing years where we started looking at it for adaptive cruise control, one of their limited applications, it was many, many years where we knew that there were significant benefits to the technology.

But getting the costs down, getting awareness of, getting the OEMs, the four forces, Xavier, that you recognized, getting traction on those was really hard. Now, we were very committed to this and part of it was our DNA as Hughes. But part of it was also a very strong belief that this market was real, it's going to happen. And now, we're seeing that actually hit. And so, we'll talk again about what are we doing from a portfolio standpoint and organizational standpoint and how are we positioned to take advantage of that.

So first when we think about our portfolio and what we do – comprehensively we call that smart vehicle architecture - and when we talk about architecture, what we really mean is how you partition software and compute around the vehicle and how you connect it all.

Historically, we always focus on the vehicle, but now, we also include off the vehicle with connectivity in addition to software, sensing and computing, signal and power distribution. And it's that combination of portfolio elements that then allow us to bring in the smart mobility solutions. Active safety that we'll talk about today, user experience, connected services, and where it all comes together is in Level 4 and Level 5 automated driving.

And really, it leverages everything we do, and when we think about our portfolio and why is Aptiv uniquely positioned when you think about what's happening in our market going forward is that we have really the full complete understanding of the brain, the nervous system and how that all works together.

From an active safety perspective; it's the software and compute which is part of the brain, and it's also the signal and power distribution, so how you move all that data around the vehicle and that is especially important as you think about next-generation systems. And I have a slide later to further articulate that. And then the foundation of security and connectivity comes into play, and then finally, where this all leads is automated driving.

And so, for us, we actually see across the entire vehicle when we look at these solutions. And what that does is it allows Aptiv to find the optimal or the unique solution for the OEM as they're thinking about Level 2, Level 3 or even higher.

On slide 34, I mentioned some of the history. Back in the 80s we were actually part of Hughes aircraft or Hughes electronics and really, again, this is the legacy that we have in the space for the first introduction of adaptive cruise control, those were all electromechanical assemblies, not unlike what you hear about with LIDAR today, very expensive, very cumbersome and not as reliable.

In 2009 though, that's when that flipped to solid state, and that was a 10-year period, very low volume, and LIDAR, we see that accelerating greatly. And then in 2014, we integrated both camera and radar into a single unit.

Up until that point in time, the world was really thinking about sensors as smart sensors, all of the compute in the sensor itself. In 2017, that's when we worked with Audi on zFAS. And the importance of that program was it was really the pivoting of the industry away from distributed smart sensors to more centralized compute with what we call satellite sensors where you take all of that heavy processing out of the sensor, you centralize that into a modular multi-domain controller and then you basically make the sensors much more easy to package, lower-cost, easier to deploy.

And then since 2017 when we introduced that, there's been a number of other new business awards that I'll talk about, that are following in that footstep. And what's driving that is that added complexities, you go from a Level 1 or really a fairly simple level of capability to a Level 2 or a 2+, at that point, you make that conversion.

So, when we look at our portfolio, one of the things that we highlight is we really have all of the ingredients that you need as you think about going from where we are today up to Level 4 or Level 5 system, it's radar, it's LIDAR, it's vision working with our partner with Mobileye and it's the multi-domain controllers, all of the software and sensor fusion, how to put all that together.

And the thing that we highlight is not only do we have all the ingredients and the portfolios, we can provide the enabling technologies, we know how to put it together and we know how to integrate the system.

And that's of tremendous value to the OEMs because as you think about the complexity of these systems, having a partner like Aptiv that actually can put it altogether, ensure that it works and launches on time with all of the features that they have committed to on those vehicle programs, it's incredibly important from an OEM perspective.

And it's separating out what we would normally think of as the Tier 1 community. There are companies like Aptiv that can do these tasks, can manage that complexity. There are a number that really simply can. There are going to be component suppliers into these systems.

So, let me talk a little bit about perception. So, radar, long history there, 25-plus years, and really, what's interesting is radar just continues to develop. I mentioned back in 2004, it flipped over to solid state. Now, we're getting into imaging radar much wider field of use and really being able to do a lot more with radar than we've ever been able to do before. In fact, you don't hear as much about radar quite frankly as we should because as sensing technology, it's incredibly robust in all weather, very insensitive to dust particulate, things that vision and LIDAR are sensitive to. So, it's an incredibly powerful sensing technology.

Vision, on the other hand, really has the strength of object classification, angular position and then non-object seeing contacts, helping us understand what's happening in the environment, and that's part of that object classification. So, vision is really critical to the overall sensing portfolio. Many, many years ago, we decided we're going to continue to partner with Mobileye and vision so that we can focus on radar, bring on new technologies and also focus on the multi-domain controller, being able to do that. So, as we thought about how do we allocate our resources, that was a key decision we've made a number of years ago.

And then finally, LIDAR, which is relatively new to the scene, starting at Level 3, you really need LIDAR, a minimum forward-looking. Level 4 and 5, you needed all 360 around the vehicle for localization, precise 3D object detection, and that's really the key for localization, being able to find objects in space with a high degree of precision, range accuracy and then something really critical for when the car is in control.

We'll refer to as free space detection because that's where the car can drive. So, the ability to understand the free space around the vehicle so you can always either pilot or navigate the car to a safe stop, that's where LIDAR really plays an important role.

Now, the reason you needed all three, because you'll hear from a lot of people, "Well, we can do it all with one or we can do it primarily with one maybe a little bit of support from the others." The reality is when you think about all the used cases that a vehicle will go through a Level 3 or a Level 4, you really need all three.

And at any one point in time, one of those will be the predominant sensor that you're relying on to help navigate the car. And you can see with the plusses and the zeroes and the minuses the relative strengths and weaknesses that the sensing modality has and it's by fusing those together that you actually come away with a really, a fully capable perception system. This is just physics. There is at the end of the day, when you're talking about a sensor, whether it's a laser or it's a camera receiver which basically is receiving or it's radar which is transmitting a microwave energy and then reflecting. At the end of the day, those physics have limitations based on the environment and there's really no way to get around those. What you need is have the right combination of sensing technologies that allows you when you fuse together to always have a robust perception system.

We would also add on top of this for Level 4 and 5 vehicle to infrastructure communication as another layer of redundancy for the vehicle to understand what's happening around it. The key is you need all three and then you need to be able to put it together and you need to be able to fuse it.

And what I mean by that is there's low level fusion where when you detect an object, you're really going to look at it and you're going to use multiple technologies to verify what that object is. You can use, for instance, LIDAR to find an object, to paint the outside of the object, vision, and to see what that object is and to do the classification, and radar to tell you how fast that object is moving towards you.

So, you need to pull all of those together so when you build your environmental model around the car, you have the highest confidence what you're looking at and what's around the vehicle.

Now, I mentioned earlier, one of the key pieces was the change in architectures. We went from Level 0 to 1 to 2 to 2+. What this shows is back in 2007, in the Volvo S80 where we basically put the level scanning radar and then an analog camera with a fusion controller. This is the complexity of the system at that time, two sensors, a 50 mega hertz clock speed relative to performance, 56 DMIPS relatively simple, less than 100,000 lines of code.

When you think about that architecture, that's really the embodiment of where the smart sensor kind of looks like. Cost optimized, trying to shrink it down as much as possible. And it was a revolutionary breakthrough at the time. I mean that was a huge step forward if you think about where we were in 2006, comparatively a massive step forward in terms of active safety technology.

Now, forward to 2018 where the work we're doing with BMW today on their systems, so, it is an active safety domain controller, so it's a centralized computer that has all of the processing basically inside of it except for some edge imaging processing.

It can support up to 16 different sensors. Clock speeds are now up above 2 gigahertz with 130,000 DMIPS and over 15 million lines of code going into it. It's a massive change in construction and content, which means that architecture of a smart sensor kind of plugging more of those simply doesn't work. So that architecture breaks down and when you come into this type of complexity with this amount of content and thinking that you want to apply it across a number of vehicle platforms with scalability across platforms, it means you go to a completely different architecture.

You go to a centralizer or a satellite sensor architecture. And where we drive the prices down, the cost down on the edge sensors, and really put all as much of the software and the content back into the centralized controller. This really shows the story of why… when you think about Level 1 and Level 2, when you think of the sensor suites that are part of those, it's fairly limited. Level 2+ is where that really begins to change significantly.

We were adding corner radar, frontal cameras, and that's where you first start seeing the Level 2 and 2+, the multi-domain controller come in. And our recent wins are reinforcing exactly this. Level 3 and above at that point in time, this is the only approach you can take. There's no tradeoff. You have to go really with this type of an architecture because now, you're talking about the vehicle being in control.

Now, when you think about content per vehicle, so what does that mean for us? It means reasonably good growth through Level 1, Level 2 and Level 2+. You see it more of a step change as you get to Level 3 though. And the reason for that is at that point, the system is in control, so it has to be fail operational, it has to have the safe stop capabilities, which means some degree of localization. So, it has a lot more sophistication in terms of what that system can do and I think you saw on Xavier's chart the big change is the driver required to be in the loop or not, is the system in control or not.

If the system is in control and all of those functional safety issues come into play, with fail operational capability and a safe stop. And today, that means a significant increase in content mainly and especially with the number of sensors, the redundancy of the sensors and then the content that goes into the central compute. The other comment I would make is as you think about the hardware costs and a big part of that is in the sensing, and when you think about LIDAR and those types of sensors, those costs are still coming down.

There's also the software license. From an Aptiv perspective, it's really important that when you think about that, all of those 15 million lines of code that we're adding where we think we have the right value. We're not going to provide all of that code. We rely on a lot of partners to provide code, but when it comes to control algorithms, planning and policy, and the sensor fusion, those are the areas that we want to play in. That's the content we want to add, not so much operating systems, middleware and sort of the infrastructure software that helps it work.

So how has it's been playing out in the market? So Audi I mentioned before with the Level 0 through 3, which was just launched here last year. That for us is a real proof point though. About five years ahead of that, we identified that trend in terms of vehicle architecture. Winning that program was important and a proof point to us.

But then since then we've been really excited about the BMW award that we announced last year. And that's really the full suite of active safety and automated driving controllers for BMW along with a number of the peripheral sensors. And that is a step function above and beyond where we were, quite frankly, was that fast. It's significantly more compute. It's significantly more capability.

Having said that, these are still premium OEMs and kind of when we talk about democratization of safety as important as these customers are in terms of kind of being at the head of the school of fish, they tend to be the ones pushing that. What's also important is really the higher volume programs.

And so what gets us to those higher volume programs, the more of the mass market? It's the things that Xavier and David both talked about. It's what are regulations doing, how are the premium or the OEMs investing and adopting, and then getting that out into the market so that the market is really pulling.

And then the four things, Xavier, I think that you mentioned, which is really important, what are the technology developers doing about bringing prices down, bringing costs down, and making the technology affordable. We're seeing, in all cases, that's really getting pushed, so all of those forces are working to broaden and deploy the technology into the market.

So we're sitting here and more of what I would call the mass market or the value market as a couple of recent wins, one with a North American OEM that basically was the scalable active safety across all of their vehicles, and then here recently the announcement with FCA, again similar type of architecture with centralized controllers, satellite sensors that easily scales across the vehicle platforms so the OEM can invest, develop and then deploy very rapidly and, in both cases, a very price affordable level for their consumers.

So this has to hit the value props that these OEMs are looking for in many of their vehicles. They're sitting right in that value segment, so really again a proof point of the democratization of these systems. We couldn't be more excited about it. To us it really reinforces the architecture discussion the fact that to be able to win these types of programs you have to have the right architecture solution that then drives the right compliment solutions.

Turning to slide 44, on China, when you think about the rest those were primarily Europe and North American-centric, but I think, Elena, as you showed in your chart and we've seen from a China perspective active safety is also now catching up. And when you think about where that market has gone from 2006 with the Cherry QQ being one of the most popular cars, no active safety really on a base model, maybe ABS is an option, to where we are today with the Tiggo 7; significantly higher contented and price vehicle with standard airbags, air-conditioning, ABS, ESP, adaptive cruise control and auto braking, it's a some massive change in direction for that market but reflects the fact that it is being pushed very heavily.

And those forces that we've talked about here in talking about here in Europe and North America with the adoption of NCAP, with consumer awareness, with the regulatory environments, movement, we expect China to really accelerate.

One of the things that we enjoy having been in China now over 25 years with a full active safety development capability in region is we can work with these local OEMs close to deploy the technologies we already have ready to go.

On slide 45, we think about the addressable market has this incredible growth projections that for the foreseeable future with a tremendous amount of upside to that. Then we look at our bookings then how do we match to that? 2017 was really a strong year for us. If you think about what we delivered in that year, this was really where all of essentially the satellite architecture came into play, 2018 already almost $1 billion through the first quarter of the year and within a projection to be north of $3 billion.

And when you think about how that translates to revenue for all those years that that I was part of active safety, when we look at what we call that the hockey stick, it's nice to see that inflection point actually hitting now and really seeing the bookings reflected and the pull-through on the revenue reflecting that as well.

We mentioned there were the strong market forces consumer adoption, OEM adoption and penetration, regulatory pressures. When you think about those it really gives us high confidence in our ability to hit these numbers or even do better. As Elena said, in 2018, we'll about $1 billion of sales.

Now lets talk a little bit about automated driving because what we've been talking about before is really the Level 0, 1, 2 and kind of in the 3 from an active safety perspective. We've made a number of investments still over the last four years in automated driving. And really at the time, I think back to when we made the investment in Ottomatika back in early 2014, the first step, it really reflected our belief that those technologies were going to be important for active safety.

At that point in time, I don't think anybody had as much clarity as we do today regarding how automated driving would unfold in terms of the automated mobility on demand and the commercial applications. We were thinking back then that we knew this was the right direction to be going.

So on the software side investing in Ottomatika, which was all of the Carnegie Mellon University IP supporting automated driving, so if you think about the DARPA challenges, Red Whittaker, the team in CMU, this was that foundation; more recently in nuTonomy, the team out of MIT. So again, original asset back to DARPA challenge capability but looking at the Level 4 and 5 automated driving.

Along the way also we know, as I mentioned, perception is incredibly important for this. And we want to make sure we had access to who we think are going to be the winners in the long run for solid-state LiDAR. So Quanergy was an initial investment, then Innoviz and LeddarTech, working closely with those companies now to make sure that they can industrialize and deploy really on the 2021 timeframe.

Originally those were more towards automated driving, but one of the things that we're looking at is not just how does our current active safety foundation help us with our automated driving development and overall efforts, but also the other way around. How does the work we're doing in automated driving help pull our work in active safety? So how do I take the technologies I'm developing with Ottomatika, and nuTonomy, and LeddarTech, and Innoviz and basically push that back into my current active safety portfolio to strengthen that? So we really look at it in both directions. It's not just building on a foundation, it's also enriching that foundation and accelerating our ability to deploy new and interesting and differentiating active safety features.

As we think about where automated is going, it's really about a connected service capability as well. Those automated driving platforms are fully connected, fully upgradeable over time. Again, how do you bring them back into our current active safety portfolio?

And then scalable architectures, as we have experienced with automated driving, one of the things that we recognize is that Level 2 forces an architecture change, and automated driving makes you completely rethink the architecture of the vehicle: redundancy, data integrity, compute, connectivity, all of those things. What that does is that gives us a clear roadmap along which we can plan for our future investments and our future capability development, right, from Level 1 to 2, to 3, to 4, to 5. So we always stay ahead of where the market is today relative to OEM development.

When you look at those bookings numbers, part of the reason for that is because we're having very different conversations with our OEM customers than we were four years ago. They see us as a systems provider, a company that can put all the pieces together, can contribute many including the content, whereas before it may have been considered more of just an integrator and a component provider. So the investments that we've made in automated driving really have changed the way we interact with our customers. It's a very different relationship now.

So to summarize, a couple items; one is we're perfectly positioned to enjoy that acceleration in the active safety market, the Level 0 through 3 positioning. And if you think about where that's going, it's not just increasing in penetration rates, it's also adding content significantly.

And I'm convinced when you think about it was a really interesting looking at, Xavier, your numbers in terms of consumer acceptance because that's consumer acceptance basically have systems that are several years old now, not the latest, not the most robust. These are some of the earlier systems, kind of like the lane departure. People have turned it off because it kept annoying them. The systems we're introducing now are much better. They're much more robust. They have much lower rate of false positives. They have a much more higher performance levels. That consumer acceptance will only continue to grow, and that will further accelerate that adoption trend.

The second is the portfolio of relevant technologies. We can help our customers differentiate starting with the Audi's and the BMWs, so the premium all the way down to the FCAs, and the Western high volume OEMs as well as players like the Chinese customers. So we're really excited about being able to offer that.

When we think about cost per vehicle, being in this space basically puts us in a very good position relative to taking advantage of the content and cost per vehicle increases. So as you saw those numbers, those are significant. The customer awards we're enjoying today, the bookings are validating the approach. The customers see value in that and we have a strong track record of winning.

And then ultimately, where we see this going is just building towards automated driving. When automated driving ultimately hits for the OEMs, that's still a good question. We know we want to get there, but it's still going to be the long time before Level 4 for the retail consumer market is introduced. Every step along the way though is beneficial for Aptiv. It really adds content and takes advantage of our portfolio.

Okay. With that, Elena, I'll turn it back to you.

QUESTIONS & ANSWERS

Elena Rosman: Great. So, hopefully that provided some context for the discussion that we're about to have. We've got about 35 minutes left on the schedule, so I think plenty of time. So does anyone want to kick us off for the first question? Maybe Brian will help.

Brian Johnson: Okay. I'll start with Xavier, the take rate on the slide, for example, you've had something like GM 20% penitration. Is that 20% of their overall U.S. sales or 20% take rate on the models with the feature available?

Xavier Mosquet: It's a great question. So this is actual take rate overall --

Brian Johnson: Overall, yes.

Xavier Mosquet: -- on the cars sold and, yes, all cars today are not available with those features.

Brian Johnson: Second and, I guess, I could have done the math, when you have the market active safety dollar billion growth numbers, is that assuming kind of price reductions of the technology that Glen talked about?

Xavier Mosquet: Yes, it's part of it, so the model actually those featuring some yearly cost reduction between now and 2022.

Brian Johnson: Okay. Now I get to the more interesting questions, probably for all three of you especially David and maybe Xavier. None of you talked about the insurance industry, and when they come into, for example, I was consulting for a large P&C insurer back when the airbags were going from standard, from available to standard.

And they actually put a big advertising effort in place to convince people to get airbags, showed very moving pictures of people saved by air bags, readily communicated discounts if you buy powered airbags, you're going to save X dollars per year, X percent on your previous.

We haven't really seen that for active safety. In fact, my insurance colleague, [Jake Galba] marching about, well, these little sensors increase the severity, we're just not convinced. Can you comment on kind of where the P&C industry is and what's it going to take within your $250 billion cost reduction number? If that was true, insurance should be all over it, yes, there is the severity issue in there, just this issue of communicated discounts.

David Strickland: I'll start, Brian. I've had lots of conversations with the Property & Casualty folks, and there's a couple of issues here. First, I think their process and their data-driven evaluations from an actuarial standpoint have changed. And I think that, as an example, I think the airbags is a good one, I think probably where you saw the last time insurance companies sort of bet on the prospect of technology is ABS.

And they didn't highly incentivize, they gave discounts but actually in the return on their investment, I think, there's a number of property and casualty companies that did not see the return that they thought from incentivizing the consumer to purchase ABS for particular discounts, and that's probably the last time you saw that type of discount.

And because this is part of the hilt on how we lost data institutes, some other folks that insurance is too for highway safety is we're working through, they actually want to really follow the data what was really an effective thing for insurance issues and make an investment.

So how does that impact for active safety? Well, first in terms of the market penetration and the initial evaluations of these systems, but actually the insurances had actually had a conversation with a number of folks about this not that long ago that it's really difficult to figure out which cars have these systems because in terms of build sheets and how you sort of break down the systems that have active safety, vehicles that have active safety, they can and it takes a lot of diving, a lot of resources and incentive for them to figure it out. So that's sort of number one. It's hard for them to figure out which vehicles. Now they've gotten better but it's harder for them to figure it out.

Number two, and I think you mentioned it, Brian, is the fact that the cost differential is much higher. When you actually had a fender bender 15 years ago it's about a $400 fix. You have a fender bender today, it could be a $3,000 fix, and the cost benefit in terms of lives and injuries, severity versus the property loss, they're having a difficult time figuring that out, too.

And in terms of the effectiveness of it as well, as I mentioned in my talk, it's much harder to prove the crash avoided. And so I think that there's a number of the insurance companies that really is sort of taking a wait-and-see and try to get better data of where they make an evaluation of whether they're going to lean in and incent consumers to buy this.

Will it happen at some point? I think it will. I just think it is a much harder process for property and casualty insurers to be able to make those types of direct investments for active safety right now.

I think it will happen eventually, but I think, as Glen stated, you're talking about older versions of these systems, which actually have gotten much better, much more robust and much more accurate. And I think that's a longer deal to catch up as well, so that's probably the reason why you're not seeing the insurance industry, as you said, jump all over this.

Xavier Mosquet: Yes, maybe I would add a couple of things. First, a few insurance companies are actually giving incentives to the customers so it's not universe wide positioned. I think David is right, they're pretty slow in doing it. But in Europe, the number of insurance companies are actually incentivizing for everything that's around pedestrian protection. We know there are incentives in the U.S. There's a few insurance companies who are giving incentives for AEB.

But the thing that they are struggling with is when the figures are low they don't know whether the customers who are buying those features are going to actually benefit from that. I mean, they're not looking at the society as a whole, but if you're, for instance, a Volvo owner and you're a very safe driver and you buy those features, you're safe already so it makes you a little safer but the value of that is extremely small. And that's the way they're looking at it. They're looking at us as individuals. And as long as the figures are not big enough to go into the population that are not to safe drivers, they don't measure the benefit yet. And so that's why they're waiting a little bit.

Glen De Vos: I will just add, I think as both David and Xavier mentioned, the data coming back from the field is still very nascent. And if you think about airbags, that was almost a digital change, the car with or without relatively easy comparison.

I think about active safety features, it's a much more complex analysis, David, as you were indicating. So I think as these systems mature as we get more experience, more data coming back from the field then you can have a much better chance of taking action on those either to incentivize or try to promote, but we're just at the very beginning of getting that data back.

Elena Rosman: Next question.

Unidentified Participant: David, can you talk about how the regulatory system has a time spin an impediment for your mostly passive features in the past? Can you talk a little bit about full autonomy and what the regulatory process will be there especially it seems like you have some OEMs who may be marketing Level 2 systems as more capable than they actually are. Is there some sort of an (inaudible) or a compliance feature that the regulators will have to adopt there and we see that play out?

David Strickland: Once again I think it's going to be a long tailed process because ultimately you're replacing the driver with a system and how do you sort of evaluate that from a Federal Motor Vehicle Safety Standards perspective because everything that you have it really is a dynamic test with particular parameters, how long does it take for your vehicle to break when you have electronics ability control, let's say here's a test and (inaudible) all these other things, it's a very specific test. So it's going to be a long time before you see the regulator having a standard for what is an automated system.

Vis-à-vis we struggle in taking out any sort of advocation of any representation of any client within the app. Probably what you're going to be looking at is probably an entirely different regulatory class. I'm not sure if people, for the folks that aren't in the (inaudible) cars, there's a low-speed vehicle class called FMVSS 500.

It's for those like carts like in senior living, neighborhoods and communities where you have your golf cart that's less than 25 miles an hour but it can actually in the public way. And the reason why you do that is that, well, the agency made a decision to carve out this class so you don't have to have the same crashworthiness standards and things around that vehicle because they're never going to be exposed to high-speed mix traffic. You're only going to be in a neighborhood community.

So for these particular vehicles, these low-speed vehicles, they have a particular structure around them, which makes them lower cost because it does provide a particular benefit. So my guess will probably end up seeing something like an FMVSS 600 for automated vehicles, which may create an entirely different class where depending upon your proofs of the ability to manage traffic, your vision technology, your compute speed, your simulation testing, all that. If you prove these particular things you don't have to have if you're a Level 4, Level 5.

You don't have to have a steering wheel, or brake pedals. You can actually have different configurations for your airbag targeting because you've met these standards internally in FMVSS 600, which puts you in a different evaluation.

That will be the logical way that the regulator would set it up, but to get to that point to create all of the simulation testing, figuring out all the other issues that need the agencies thought about in its initial guidance and much more, it's going to take a long time before it gets there.

Elena Rosman: Next question? If I could, just building on the last question, and the marketing piece of it, David, can you comment? We've had a number of questions around the marketing of Level 2, Level 2 plus systems, and whether that's influencing the level of confidence consumers have in autonomous driving.

David Strickland: Really Level 2 and Level 2 plus are the foundational steps to getting to Level 4 and Level 5. And I think what the agency has been looking out for a number of years when I was administrator, the past administrators and the current team, is that there are so many lives that we saved by the utilization of Level 2 technologies, Level 2 plus technologies, getting those fully penetrated.

You're talking about thousands of potential lives. When Xavier talked about this on his slide deck is that you don't want to jump these steps because as you're getting towards full deployment, which is we all mentioned, it's a long way out when you get to an individually-owned Level 4 vehicle.

You want to get these technologies into the fleet as quickly and as deeply as possible because those are all crashes avoided. So I think in terms of how you think about from a governmental standpoint, a societal standpoint and even the safety groups are talking about this as well that really the exploitation of Level 2 and 2 plus, I think, is very important.

And circling back into the marketing aspect as well, Elena, in terms of pressures on what is a safe vehicle in model year 2025, 2026. As you get deeper penetrations of active safety systems, comparatively speaking to manufacturers that build vehicles without them or without them being fully democratized like Toyota is probably the best aspect of it.

Toyota could end up being a real market mover for the entire industry because as they push down their active safety suite down to their entry-level vehicles, there is going to be a long hard conversation about folks, you know, compared to, A, Toyota, for the same price, provides these systems or for a minimal cost increase compared to others in that segment cost increase all the way to an enterprising trial or saying, well, you know something, all these manufacturers have these systems.

I would think that it is unreasonable proposition for you, Manufacturer X, to not have this system in a new vehicle and you have a higher risk of crashes, higher risk of injuries. And because your system was on my client's vehicle, this crash could have been avoided but it wasn't. So therefore, we're going to try to hold you liable.

So all of those things together, I think, is that push to making sure the Level 2 and 2 plus, I think, are getting much more deeply penetrated through all manufacturers before we sort of see the broader adoption of these higher levels, which are going to be much more difficult to implement.

Elena Rosman: Thanks. Dan?

Dan Levy: Thank you. I just want to follow-up on that in a slightly different vein. Just give us a sense from the perspective of the OEMs, to what extent is Level 4 driving and development of those system being done by the same teams or the same groups as the teams that are implementing Level 2 technologies. Are there different pushes within the OEMs that are looking at it from a different lens?

Glen De Vos: I can certainly provide this. Xavier, you probably have thoughts on it as well. It really varies quite a bit. If you think about the path GM or Ford have taken - Cruise in the case of GM, Argo in the case of Ford - they've gone with fairly autonomous groups that are doing Level 4 and 5. They basically have a breakpoint.

And typically the ADAS or the Active Safety teams that are inside the mainstream organization are responsible up to, and sometimes including Level 3, there's a bit of a crossover that occurs there. Others, as we work with some of the European OEMs, they are maybe separate groups but within the same departments.

So if you think about like BMW, for example, has put together their tech center in Unterschleißheim, which is really their automated driving center. And part of that is working on Level 2 systems but part of it is also 3 and then looking at 4, but they have different areas within that overall organizations by 1,000 people there.

And part of that reflects that Level 4 and Level 3 to a great extent, those are robotics challenges as opposed to Level 0 through 2, which tend to be controls challenges, so you have to think about it a little bit differently as they have a philosophy associated with how you solve those two problems.

One is with the cars and controls so I think of that more as robotics challenge, which is a different skill set than the traditional controls engineers that have been developing ADAS systems. And so that naturally kind of bifurcates the organization. But in the case of BMW, those are kind of inside of the same groups under the same overall management teams to make sure that they're working closely together. And then we, as a partner, work across all of it.

And then I would say Asia is a little bit different as well in terms of how they're set-up, so you get the full variety.

Dan Levy: So then within Aptiv, how much, and I think everything is reporting up to you, but how much is their data or input or something from whether it's nuTonomy or Ottomatika, your Level 3, 4 systems that are inputting the system that you're shipping out of the development of the systems that are commercially being shipped out today where you're getting that billion dollars of revenue today. How separate or --

Glen De Vos: So we separated out the organization, so under me is part of the Mobility and Services Group is that's where we put the nuTonomy and the Ottomatika teams and some of the Aptiv engineers as well to really focus on the commercial applications because those customers like Lyft and other ride-hailing networks are very, very different than the OEM customers.

And then within active safety and user experience, under David sits what we call AD OEM group. That's the group that interfaces with the OEMs and they're more traditional organizations. Now, that's kind of the customer-facing piece of it. On the technology side, we have quite a bit of collaboration because if you think about what I'm doing on commercial and automated driving, I'm trying to leverage all the radar, camera work, all of the work that's already been done basically by the active safety organization and what they are developing. And I'm trying to give them roadmaps to support me, and then vice versa.

As they work with OEM customers, they're quickly drawing on the Karl's and the Emilio's at nuTonomy or Ottomatika to provide them with guidance and understanding of how these Level 4 and 5 systems really are going to work. So there's a lot of collaboration kind of below the customer-facing piece of it, but we have made a conscious decision to make sure so that we can best serve those markets to break it apart from a customer-facing standpoint.

Elena Rosman: Elliot

Unidentified Participant: I just wanted to ask maybe a generation of the question that's already been in place, which is so what are the assumptions you're using in terms of cost downs? It seems like they would be because it's tech-related it's going to be more aggressive than a standard OEM whatever call it 1% to 3% a year.

And then specifically, beyond that, I guess, are there certain tipping points that you see where, hey, we can get a level X system below Y price? That's when adoptions really going to spike or is it going to be more regulatory or driven from the OEM side?

Elena Rosman: Maybe I can start and then ask both Xavier and Glen to comment. Xavier touched on this earlier, but the cost of the sensor suite has already come down to a point which it can drive the active safety penetration we're talking about today.

Going forward, we typically see mid single-digit average selling price declines, factored into our outlook. Separately, there's an element of software abstraction, integration and complexity that would be carved out of that analysis. Glen, perhaps you could provide further perspective.

Glen De Vos: Sure, maybe I'll start. In favor of Level 4, a couple of years ago as we looked at the market, we judge that for mass adoption we needed to be right about that $10,000 of on cost to the end consumer for the system. And that number in terms of what the consumer is willing to pay at least with what they currently understand the system can do seems to be whole so that kind of translates to I got to get the system down to about $5,000 of initial cost back to the OEM.

Now to get there, so that would kind of answer what is that big trigger point. Now to get there, there are a couple, when you think about compute, when you think about cameras, radars, those types of devices, those are fairly mature. There's a lot of discussions on Moore's Laws kind of running out of gas and so you're not going to see massive declines year-over-year in those technologies basically. Memory prices, things like that are fairly stable and we continue to optimize, but it's not a big step function.

The one exception to that is LiDAR where today there's not a solid-state LiDAR available. And when they first come out so like what we saw with radar to this 10 years we get to a solid-state radar, it's not going to take that long with LiDAR fortunately, but there were massive declines.

At first radar wasn't many thousands of dollars compared to $150 today. So you are going to see LiDAR come down dramatically. And because we use it for automated driving in a 360 fashion on the vehicle, that's a big lever, so when that has to happen early in deployment, we think that will happen in the '21, '22 timeframe, somewhere in there so that when we think about OEM getting to those price points is post '25. That's roughly how we think about for the retail consumer markets.

Commercial can withstand the economics of a much higher level of content though, which is why we also want to play in that space near-term since that can afford a much higher content per vehicle level than the retail consumer markets. But that's kind of how we see that. Xavier, I don't know if you want to add.

Xavier Mosquet: I would like to add, if you look at historically with active safety systems were, on average, twice as expensive as what the average consumer wanted to pay. And so we have to increase scale and this is why the fact that OEMs have made commitments to raise the scale and making it more standard allows us to reduce complexity and increase the volume at a point where it's thought to be affordable.

Historically, you look in the past, whatever six, seven years, the cost of price decline has been somewhere between 3% and 6% depending on the features. That will slow down a little bit because now we're sometimes at 20%, 30%, 40% penetration rate, and so that cost decline will sort of stabilize a little bit, but you're expecting this is electronics in a way, so expect to actually those costs to continue slowly declining in the future except for these people, the newer types of centers that Glen was talking about where it was starting at the top of the curve, and so the next 10 years are going to see significant cost decline.

But I agree that we have to be able the volumes for commercial applications and decisioning. Somewhere the commercial applications in the premium OEMs are going to be the ones paving the way for L4, L5 technology. And then within five, six, seven years, then the cost we'll make it more affordable and to more volume types of customers.

Glen De Vos: There's also something else I would add, and that is the revenue models are going to change. If you think about revenue models for us today it's a unit sale up front and then we never see the product again or we hope to never see the product again.

If you think about these systems it should be learning systems, so there's going to be an ongoing maintenance in terms of the software, of the service level agreements for software maintenance, continuing to improve and basically extend the number use cases that a given vehicle post sale can manage.

So to do that means, and this is why it was so important for us to have content and policy planning and that piece of that software, to do that means there's going to be ongoing engineering and support activities for those vehicle long after that sale is initiated. So that means that's an opportunity for recurring revenue coming out the vehicle.

Now that's a very different model than what we see today. And like I said, that was one of the reasons why it was so important that we have software content on that vehicle and critical content because, as we make that better, as you learn from who did it coming back from the fleet, we want to continue to upgrade that vehicle. Now those models we're still trying to figure all that out, but we know that potential is there. We certainly want to take advantage of it.

And if you think about how that can be consumed, 2025 is a ways away. If you think about consumer preferences or how that can be consumed, that could be a way to reduce that initial cost back to the end consumer making it back up to recurring revenue or consumption-based revenue models. That's one of the exciting parts about the changes that are occurring in our industry.

And again I'll emphasize, for Aptiv, why it was important for us to be more than just a component supplier or more than just a systems integrator where basically you do the development, you're done and now you're getting basically a unit sale environment.

Elena Rosman: Next question?

Unidentified Participant: Thanks. While we're talking about software, Glen, you had on Slide 40 the CPV for Level 2 plus and Level 3, and the sensing and compute as of the CPV but then you have the software element, can you just talk to what potential we could add to the CPV for including the software?

Glen De Vos: Yes, if you think about it for the commercial market it's substantial. It's roughly equivalent to the hardware cost in terms of getting the deployment. And that's because that's an economic model, and what we're looking at is essentially what value can you capture as you take out the driver, and you enable the automated service. And so we understand here's the hardware costs to do that. That gives us evaluation for what we think the software is worth to do that, and that's what we want to maintain.

And so when you think about the amount of tech that can go into a robo-taxi, for instance, generally speaking it has to be when you figure out 2021 timeframe for those deployments, more scaled deployment not the small fleets but bigger fleets, that car can be $125,000, $150,000 roughly. And so if you're starting with a base car maybe at the 70's, that's a lot of content that you can add. And a significant portion of that can be the software license and the software support cost that go with that.

As you go to the retail OEM market, that's a much, much more price-sensitive market but a much higher volume market. You know, you're talking about much, much greater scale, so typically in terms of software licenses, now you're thinking more in terms of kind of more traditional models where the software license may be somewhere between 10% and 20% of the value system, and then along with the service level agreement to support that. That's part of what we're working with the OEMs now to sort out where is that sweet spot.

Unidentified Participant: So just a quick follow-up on the pricing of the consumers is a Level 2, it's a question for David following up on the Tort law implications of this. Let's say that, you know, some OEMs make it standard, other OEMs having available at a modest cost about $600 on a $25,000 vehicle, but an OEM chooses to price it at like $3,000 or $4,000. Does that expose that third OEM to civil liabilities in terms of you price this feature too high to prevent the accident? I don't know if there's any precedent in motor vehicles…

David Strickland: No, I mean, I think ultimately the question is going to be because of that particular juncture the argument would be that it's a question of contract of adhesion or not. If a manufacturer has that particular system available but the consumer chose not to do it, that is the consumer's choice, and the consumer basically, it's not the same thing, it's an assumption of risk factor but it is.

I mean, it's like, look, yes, I could have chosen to buy this vehicle but I didn't want to spend that additional amount of money for it, so I assume the risk that I'm going to take on the driving task on my own, I would like to think are adaptive systems.

Where you get this tort pushed is whether it's a binary one. We have nothing. If our entire fleet doesn't have these systems versus everybody else's fleet, who does? And that's how the torts will incent that laggard to then at least make it available. It may not make it mandatory or standard, but it'll make it available.

Unidentified Participant: A question for Glen. You talked about obviously LiDAR, Level 4, Level 5. You said Level 3 as well. But if there really are breakthroughs in solid-state LiDAR, would it just become part of Level 2, Level 2 plus systems for basic AEB functions? What use cases, let's assume it was $5,000. Would it open that would improve the functioning in those systems?

Glen De Vos: Yes, as LiDAR cost come down, this is the one thing that's true about engineers, we tend to proliferate sensors. So we want to use more. We never go the other way. We never take sensors out, so there's a lot of discussion. Sometimes all the levels here can reduce the sensors and use fewer. We tend to drive sensor cost down and use more. I don't know of any example that's done the other way.

Now, having said that, we're also driving down into the Level 2 systems because then you can start adding things. There's really two benefits. The first is, well, you really want to get to an all-weather capability, all-environment capability, which many of these systems today simply don't have.

If you're driving in the worst weather, they turn off. When the system is facing the sun, if it's vision-based, it turns off. And so by adding LiDAR or other sensors for that matter into the Level 1 and 2 systems, you give them much greater capability across all the different use cases and for those basic functions.

The other is you have the ability to do more interesting things. If you think about LiDAR with its free space detection, then you can think about low-cost accident avoidance, so the ability to try to avoid something as opposed to just hitting the breaks because you can map the area out around it, you know the free space so you know where the vehicle could potentially go. And that gives you, in essence, more feature capability as well. So those are the two dimensions that you can look at.

And then the market forces that will drive that is if you think about today when you pick up your consumer reports, there really aren't any evaluation criteria comparing one active safety system versus another, I mean, by and large. As this market matures, you're going to see more differentiation and more granular differentiation of what these systems can do. And so the system that can do more of these things can be safer. I mean, it's going to be a differentiator for that OEM.

And the OEMs recognizing these sells and this differentiates we'll be looking at how to do that, and that's what we get to show them. We can show, look, if you put a LiDAR on the cars you can do this type of a thing. If you have it in front, you can do something different. And so we're anxiously anticipating getting those prices down so we can actually use more of those sensors.

David Strickland: I just want to say, is I want to talk about the engineering and marketing disruptors, which is your business development people and your lawyers, which is then you have intrepid engineers that say, "Hey, look, as opposed to let's get the cost down (inaudible) more sensors, you're going to have the business development," and your lawyers will say like, "Well, do we really need to do that? Maybe we can use few of these systems that are the same thing." Why once again the whole school of fish theory?

Best-in-class is good, but we can actually save an X amount of dollars per unit if we shrink/consolidate but get the same account, the same amount of performance, other things that can also push the department out as much as is also fuel economy standards and other things.

Every ounce of weight you can shave off of a vehicle these days folks want to do it, so if you could you use fewer systems, fewer components with the same amount of power so you can have, so while I totally agree with Glen, engineers have a way of doing that but there's other forces that may create an opposite proposition.

Unidentified Participant: Which gets to another question. Can you talk a bit about the competitive landscape really around Level 1, Level 2, Level 2 plus? How long (inaudible) kind of first the Mobileye ecosystem versus people doing their own vision. Some of the Europeans like Conti and Veoneer.

And then secondly, within the kind of Mobileye partner ecosystem, because a lot of Tier 1s will stand up and say we've got a great growing relationship with Mobileye. So what are the competitive trends there and what was the meaning for the pricing in terms of our OEMs getting to the point where they can play different Tier 1s against each other?

Glen De Vos: So in terms of the competition kind of if you break the world up into the Mobileye and non-Mobileye crews, at the end of the day, some of the non-Mobileye have close relationships with OEMs that basically help solidify their position there.

But having said that, generally speaking Mobileye still enjoys an industry lead because of their scale, the experience and the fact that as a functional safety system it's incredibly sticky for an OEM to move away from a Mobileye solution or a Mobileye architecture is a huge decision.

I mean, that's a big change in direction for them. And until there are a number of proven alternatives they typically don't. And I would say, among the Bosch [counties], again there are the others, Mobileye still enjoys a significant advantage in overall performance.

Now having said that, the market is developing. And one of the things that, you know, the playing field is leveling out a little bit. And then the reason I say that is because with advanced GPUs with more basically machine learning and A.I. tools at developers' fingertips, alternative systems are being developed and being promoted, and they're advancing very rapidly.

If you think about it even with Mobileye, they're relatively new into the A.I. space as is most of the industry. So that's actually an accelerant for broader development of vision systems that the automotive space really have enjoyed before. And along with that, you're seeing a lot of talent being attracted into automotive in the vision and perception systems.

So the landscape is changing rapidly for Mobileye where they used to be quite honestly, unless you did it yourself, they were the only option. I mean, that's encouraging for the industry in general.

Now, with regard to the Mobileye partners crew, I mentioned earlier we made a very important decision a number of years back, 14 years ago. Look, we're going to partner on vision. We're not going to try to invest and develop our own. Where we're going to go deep is in radar and where we're going to go deep in compute and multi-domain controllers and systems integration and software development.

And so what we're seeing there is that when you think about the other Mobileye partners like Valeo, TRW ZF, some of the others, we have, I think, significantly greater capability there so we can offer a much stronger value prop to the OEM across more than just one sensing modality.

We've offered a complete system and are able to show here's how you develop and implement that Level 1, 2, 3 and also (inaudible) type of system. We just don't see or there's Magna or the others having that capability. And so that's a big part of why when we think about the wins that we've had, we've been able to deliver those.

Elena Rosman: Maybe just to put some numbers behind that, Brian, so of our active safety business, which was $600 million last year in revenue, $3.7 billion in bookings, roughly half of our revenue and bookings are accounted for by radar, short-range, medium-range, long-range radar.

We also continue to win forward-looking camera systems with Mobileye for vision processing. There we won 16 platforms in 2017 across six different OEMs.

Next question, in the second row.

Unidentified Participant: You had talked about your progress with some of the North American OEMs. Xavier mentioned that the European regulators are pushing more aggressively. Would you talk about progress with some of the mass market European OEMs?

Glen De Vos: Yes. So we launched on the Scenic with Renault-Nissan in Europe here, I guess, it was last year. And that was an important program for us because it was really taking kind of a RACam-based technology and moving that into a different customer.

And that program has gone exceptionally well, and so for us that was a great foundational development activity with them that we continue to build on. FCA, by extension has also been a very important customer in North America but also extending to Europe.

And then what I would add to that is we continue to see making inroads into some of the other higher volume. And this is a great example where if you think about our portfolio we do a tremendous amount of body electronics work. Basically, body controllers are multi-domain controllers and with PSA and some of the others, so we're using that as a platform now to work into those customers as well.

Elena Rosman: Next question.

Unidentified Participant: Thanks. I got two regulatory questions, one, it's following up a little bit of question earlier like a Tesla and all the issues that they have had and how they kind of promote their auto-pilot. We've seen the Nissan, the commercials where they decided to put on their profile, right between two 18-wheelers, which probably isn't the best time to activate system. How is kind of regulation kind of thinking about how these things are promoted?

And two, look back, from your example, 10 to 12 years to get fully regulated in things, do you think North America will have sort of a regulated L4 or L5 system like it has to have these sensors and this suite and this processing unit and things like that?

David Strickland: First, in terms of its issue in terms of how you market or advertise this on this jurisdiction, safety jurisdiction, so it's literally whether a vehicle, if it's involved in a crash or crash scenario, did that vehicle's performance create an unreasonable risk to safety?

What you'll probably talk about is more of a Federal Trade Commission evaluation that, yes, like Tesla calls its system auto-pilot, but it really isn't an auto-pilot. I mean, if you think about the true definition of an auto-pilot, that's really more of a notion of deception more so in the marketplace.

And I don't think that at this particular point that it doesn't seem that the Federal Trade Commission has an appetite for whether or not that auto-pilot is a deceptive description of a particular system because there's some aspects of it that, in the right conditions, you are able to drive hands off of wheel much like a pilot does even though for short periods of time.

So what I think what will probably likely will happen in this space more so secondarily are things like Tesla is that there's probably going to be an evaluation of where there is foreseeable use and abuse. And I think that's really going to be where NHTSA will then make a decision that you did not do enough to foreclose a driver to abuse the system.

In the very beginning I made a very (inaudible), internally one thing a former administrator always do you try not to backseat administrator another administrator. When Tesla had those first videos, not the Tesla, but Tesla always had those sort of videos of people jumping in the backseat with auto-pilot engaged on YouTube, that's screaming of foreseeable use and abuse, and you really need to enforce against Tesla to make sure that they made the system more robust so it can get abused that way.

Well, Tesla on its own has improved the system because of the crashes that have happened since then. But I had a colleague who might show me a video the other day of the pressure sensor they used in the steering wheel to make sure that auto-pilot disengages if the person's hand is off the wheel for a certain number of [sections].

Well, actually there's a software thing you could buy aftermarket called Autopilot Buddy, which will take off the system, so it allow auto-pilot to drive without actually having to hold onto the wheel. But there's actually even more intrepid person on YouTube that figured out that you can wedge an orange into the crux of the steering wheel for Autopilot Buddy to (inaudible) $0.65, wedge it in there and the vehicle will not engage its reminder system where it won't deactivate the system if you don't reengage with it, so there's lots of things that are going on there in terms of how you sort of think about that. That may be how NHTSA really sort of thinks about it engages. And at Level 2 is how do you make sure that the system can't be simply circumvented and therefore take care of safety.

On your second part of your question, Level 4, Level 5, you're going to see the market create Level 4, Level 5 systems first and you'll see how those systems operate. The agency will be very close conversations with those manufacturers at Level 4, Level 5. And after they have enough data then they'll begin the process of rule. So it's just like active safety.

There's no such thing as having a premarket regulation ready to go before something comes into place. So you'll see, frankly, without there being a regulation for a long period of time, the Level 4, Level 5 will be regulated by the broadest notion of does that Level 4, Level 5 vehicle, if it's in a crash scenario, does it pose an unreasonable risk of safety. It'll use the catch hold authority versus having a specific regulatory design standard.

Elena Rosman: Great. Next question.

Unidentified Participant: Question for Xavier, I know that electric cars and active safety are not mutually exclusive, but from your research in terms of appetite for consumer to pay up, how do consumers prioritize spending on full electric versus paying up for the extra maybe Level 3.

Xavier Mosquet: Yes. So I think it's a great question in the sense that you're the consumer and, at some point, you will not have unlimited resources to spend, and there's a few things. First of all, if you look at the future of electrification, we think that somewhere between, let's say, 2023 and 2027, there will be a true economic proposition for the customer to buy electric cars.

At the speed at which your battery costs are declining, we're pretty close to the point where, frankly, as a buyer you and I will look at what's the payback in the next two years and we will be buying an electric vehicle.

It will be even more true if we're high mileage types of vehicles, so if you're a taxi or a Uber driver, I mean, this is within three years but actually the best proposition will be an electric are anyway. So, whether it's expensive or not, this is going to be the things you got to do.

Now, at the same time, and we talked a little bit about the content per vehicle of the various active safety features, we still feel that the threshold (inaudible) as long as you are within the Level 2, 2 plus, I think you are within something that is close to what a normal owner of a car would spend and the value is definitely there.

When you start getting to Level 3, there is a significant threshold, and I think at this stage there's going to be a real competition between power train and your way safety or convenience features are. Before that, I think they would go together. After, I think only the premium owners at least in the first three, four, five years will be able to afford it.

Elena Rosman: Mike, do you also have a question?

Unidentified Participant: Thanks. I think you make a good case for the technologies that are sampled in the portfolio and the underlying linkage to the electrical architecture. But from where we sit in the bleachers where there's sort of a wave after wave of press release from various companies participating and whether it's mobility services or automated driving or safety systems, it's difficult to really understand where different players are.

So rather than asking you to sort of criticize or critique competitors, maybe there are other things that you might suggest we can watch other than the bookings, which are impressive, or maybe there's other things that you watch for monitoring your own positioning in sort of a development of the technology and the commercialization.

Glen De Vos: I think I would echo your point, it is really complex and it's hard to kind of filter out. It's hard to filter out all of the kind of noise that comes through the system. I'll talk about the things that we watch.

First of all, and I'll start with the automated driving piece of it. The first thing we look at there is who's deploying at reasonable scale, and what I mean by that is who's operating fleets, meaningfully-sized fleets that are doing something, not just demos but actually performing a service. And so as we look at the automated driving landscape, I mean, that's a pretty small group. Quite frankly, that's Waymo, its crews. It will be Uber again shortly, I think, and it's Aptiv.

And all of us are deploying vehicles in, I would say, commercially focused operations, and that's what we're doing in Vegas. And the whole point of that activity is for us to understand, you know, as our role is to sell enabling technology into ride-hailing networks on the commercial side, how do we do that best, how do we price it, how do we operationalize it so that we can really take advantage of that uptick in 2021, '22 as that really ticks up.

So that's the first thing we look at, who's really doing meaningful deployments because you'll see lots of announcements about people driving in a city or people demonstrating something on 101 or in a different area. But that's kind of like the first couple of baby steps you take so you have to get to a meaningful fleet deployment for really in what I would call the major leagues.

The second thing we always look at is can people clearly articulate their path to commercialization and how they're going to commercialize the technology. And that's true whether that's a Level 4, 5 system or a Level 1 or 2 or 3. And that's why it's important for us to really be clear about how we're thinking about automated driving deployment.

We don't want to be a fleet operator. We don't want to be a network operator, a fleet management company or a bunch of cars, we want to provide technologies that enable those vehicles to plug into a network, and so we have a ecosystem of partners that we can talk about.

We have a business model that we can talk about. We have deployment targets now that we can start talking about. And then we'll talk about a lot more of that over the next six months, but that clarity around your commercial strategy is one that we think is really important as well.

And then the third thing is we always look at do people really have a right to play in the markets that they're talking about playing? What I mean by that is do you have the capability, the resources, the knowledge, the assets to actually be a meaningful player in that space. And it's one thing because we hear a lot about our technology provider is talking about automated right driving, but the reality they just provide a piece of it.

Yes, they can do a lot with that piece but they can't deliver full systems. And that's a big differentiator. I mean, it's wanting to say, "Look, I'm going to have a chip or I'm going to have an algorithm." It's not a thing to say I can actually put the whole system in place that enables the car to be an automated vehicle, and I can help an OEM to do that. You don't have to go find a bunch other people, I can do all of that for them. And so that right to play piece is another area that we look at carefully.

And that applies whether it's a Level 4 system or a 3 or a 2 or a 1. If you think about the lower level systems, the differentiator there is who can really do the systems integration the way we think we can for the OEM. Who can provide that content stitched in together? That's kind of how we look at it.

And I would say we do spend a lot of time on the competitive landscape because it is a dynamic environment and things change almost every week. It seems like something there's another announcement of something from somebody.

Open source platforms in China, new pilots in Israel, I mean, it's just every week there's or something interesting. But that's kind of how we go back to, Okay, those kind of three basis premises (inaudible) are they really deploying in a meaningful way? Do they have clarity around the commercial model? And do they have a right to play in the spaces that they're claiming?

Elena Rosman: I don't want to miss anyone, if anyone else does have a question that they want to ask, please feel free to raise your hand now. Otherwise, I think that's a good place to end.

So that concludes today's formal presentation and our informative panel discussion. I want to thank you for your participation today, including our speakers, Xavier, David and Glen. I'd like to also thank our hosts, Barclays - Brian Johnson, Dan Levy and the Barclays team.

I now would like to invite you to join us for lunch, where we can continue our discussion. Thank you again.